

6(a)(i) Force = $qE = 1.4 \times 10^{-15} (3.0 \times 10^3) = 4.2 \times 10^{-12} \text{ N}$

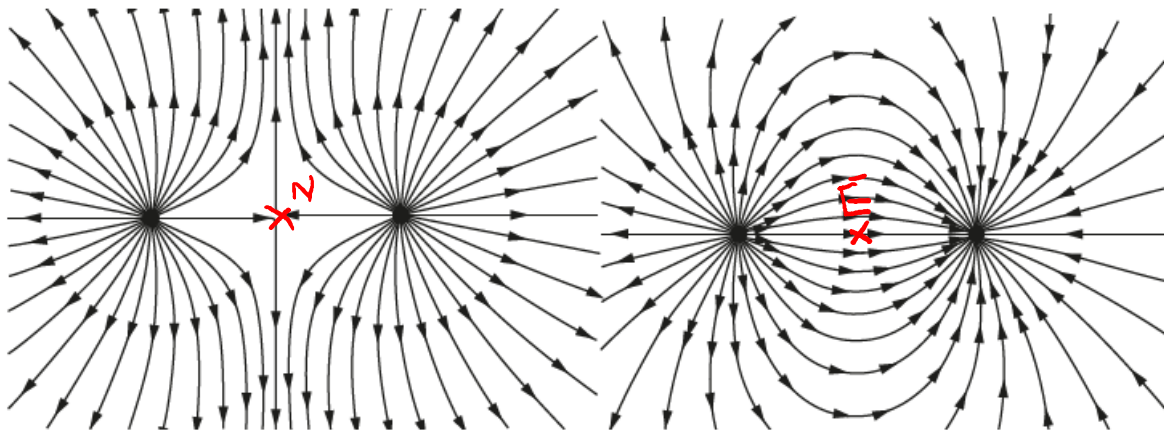
(ii)

A couple acts on the particle since two parallel forces of equal magnitude but in opposite directions acts on the particle, i.e positive charge experience a force to the left and the negative charge experience a force to the right.

The torque of the couple causes the particle to rotates anti-clockwise to align with the field. OR

The torque of the couple causes the particle oscillates about a position with the positive charge in the direction of electric field.

(b)(i)



(ii)

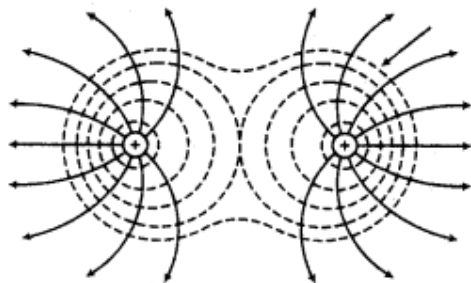
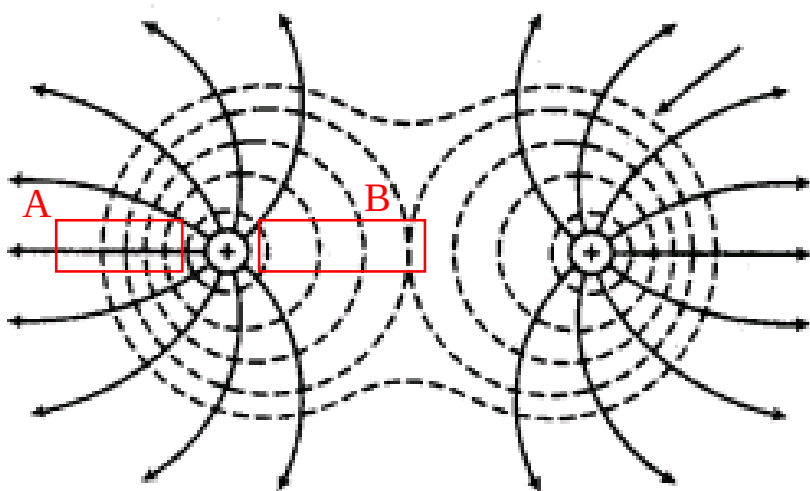


Fig 6.2: Dotted lines represent the equipotential lines.

- 3 equipotential (labelled) of roughly correct shape (lines must be perpendicular to the field lines)
- Spacing between each line is difference.

Case 1: 3 circles drawn on the same charge

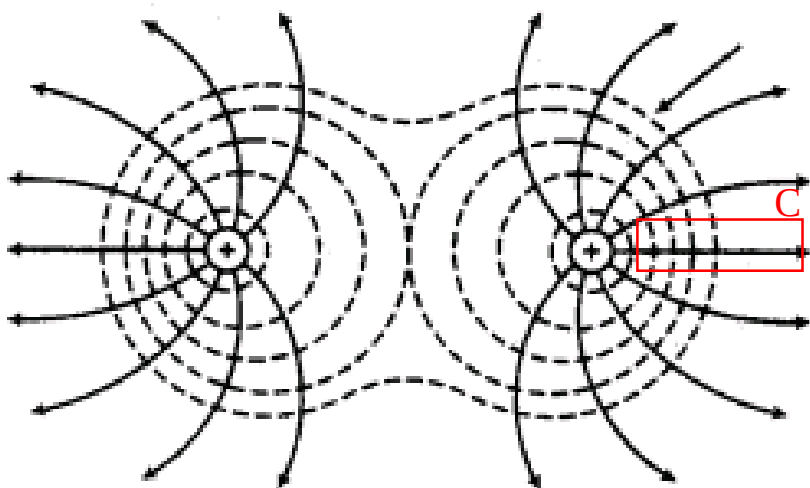
See red rectangles A and B to check for spacing between each line. In rectangle A, spacing between each line < spacing between each line in rectangle B.



Case 2: 1 circle drawn on the same charge with 2 complete lines drawn surrounding both charges. Or

Case 3: 2 circles drawn on the same charge and 1 complete line drawn surrounding both charges. Or

See red rectangle C to check for spacing between each line. In rectangle C, spacing between each lines get wider.



6(c) Student's answer must include the following 4 points:

1) Area under E(R) electric field strength graph

- Area under E(R) field graph between 2 points gives the change in potential between the 2 points.

From E(R) graph, change in potential can be calculated using $= 1 \times 15 \text{sq} (4 \times 10^6 \text{ J per square})$

$$= 60 \text{ MJ}$$

From V-R graph, change in potential is from -90 MV to -30 MV.

2) Gradient of V(R) potential graph

- Electric field strength E is the negative potential gradient (OR negative gradient of V(R))
i.e $E = -dV/dR$ (MUST write in words)
- From the V-R graph, draw 1 tangent at $R = 100$ m. gradient (MUST show working).

Gradient = magnitude of the E-field strength = $100 \times 10^{-4} \text{ Vm}^{-1}$

From E-R, at $R = 100\text{m}$, the magnitude E-field strength is $100 \times 10^{-4} \text{ Vm}^{-1}$